PHY 614 : Quantum Mechanics

Problem Set No. 1 : **Due : September 09 ,2011**

Solutions to all the problems will be posted on the website after the due date.

1. (20 points) The lagrangian for a charged particle restricted to move on a two dimensional plane with coordinates x and y in the presence of a magnetic field in the z direction perpendicular to the plane is given by

$$L_1 = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) + \frac{eB}{2}(x\dot{y} - y\dot{x}) \tag{1}$$

   (a) Calculate the conjugate momenta.

   (b) Calculate the Poisson bracket $\{\dot{x}, \dot{y}\}_{PB}$.

   (c) Calculate the hamiltonian H_1 which follows from L_1.

   (d) Obtain the equations of motion which follow from L_1.

   (e) Now consider a different lagrangian

$$L_2 = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) + eB(x\dot{y}) \tag{2}$$

   Obtain the equations of motion which follow from L_2

   (f) Calculate the hamiltonian which follows from L_2

   (g) Obtain the equations of motion which follow from the hamiltonians H_1 and H_2

   (h) Explain why the equations of motion which follow from L_1 and L_2 are identical.

2. (10 points) Exercise (2.7.1) of Shankar. (Ignore the last sentence "*Note the similarity* $\cdots$)

3. (10 points) Exercise (2.2.4) of Shankar

4. (10 points) Exercise (2.7.8) of Shankar

5. (10 points) Exercise (2.8.2) of Shankar

6. (10 points) Exercise (2.8.3) of Shankar

7. (10 points) Exercise (2.8.4) of Shankar

8. (10 points) Exercise (2.8.7) of Shankar

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Problem 1

$$L_1 = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) + \frac{eB}{2}(x\dot{y} - y\dot{x}) \quad (7)$$

(a)
$$P_x = m\dot{x} - \frac{eB}{2}y$$
$$P_y = m\dot{y} + \frac{eB}{2}x \quad (8)$$

(b)
$$\dot{x} = \frac{1}{m}\left(P_x + \frac{eB}{2}y\right)$$
$$\dot{y} = \frac{1}{m}\left(P_y - \frac{eB}{2}x\right) \quad (9)$$

$$\{\dot{x}, \dot{y}\}_{PB} = \frac{1}{m}\left\{P_x + \frac{eB}{2}y, P_y - \frac{eB}{2}x\right\}$$
$$= -\frac{eB}{2m}\{P_x, x\} + \frac{eB}{2m}\{y, P_y\}$$
$$= -\frac{eB}{2m}(-1) + \frac{eB}{2m}(1)$$
$$= \frac{eB}{m} \quad (10)$$

(c)
$$H_1 = P_x\dot{x} + P_y\dot{y} - L$$
$$= \frac{P_x}{m}\left(P_x + \frac{eB}{2}y\right) + \frac{P_y}{m}\left(P_y - \frac{eB}{2}x\right)$$
$$- \frac{eB}{2m}x\left(P_y - \frac{eB}{2}x\right)$$
$$+ \frac{eB}{2m}y\left(P_x + \frac{eB}{2}y\right)$$
$$- \frac{1}{2m}\left(P_x + \frac{eB}{2}y\right)^2$$
$$- \frac{1}{2m}\left(P_y - \frac{eB}{2}x\right)^2$$

~~Expa~~ This leads to

$$\boxed{H_1 = \frac{1}{2m}\left(P_x + \frac{eB}{2}y\right)^2 + \frac{1}{2m}\left(P_y - \frac{eB}{2}x\right)^2} \quad (11)$$

(d) Equations of motion from Lagrangian

$$\frac{dP_x}{dt} = \frac{\partial L}{\partial x} \Rightarrow \frac{d}{dt}\left(m\dot{x} - \frac{eB}{2}y\right) = +\frac{eB}{2}\dot{y}$$

or

$$\boxed{m\frac{d^2x}{dt^2} = eB\dot{y}} \quad (12)$$

$$\frac{dP_y}{dt} = \frac{\partial L}{\partial y} \Rightarrow \frac{d}{dt}\left(m\dot{y} + \frac{eB}{2}x\right) = -\frac{eB}{2}\dot{x}$$

$$\boxed{m\frac{d^2y}{dt^2} = -eB\dot{x}} \quad (13)$$

(e) For Lagrangian

$$L_2 = \frac{m}{2}\left(\dot{x}^2 + \dot{y}^2\right) + eBx\dot{y} \quad (14)$$

$$P_x = m\dot{x}$$
$$P_y = m\dot{y} + eBx \quad (15)$$

Thus eqns of motion are

$$\frac{dP_x}{dt} = \frac{\partial L}{\partial x} \Rightarrow m\frac{d^2x}{dt^2} = eB\dot{y} \quad (16)$$

$$\frac{dP_y}{dt} = \frac{\partial L}{\partial y} \Rightarrow m\frac{d^2y}{dt^2} = -eB\dot{x}$$

(f) The Hamiltonian is now

$$H_2 = \dot{x} P_x + \dot{y} P_y - L$$

$$= \cancel{m\dot{x}^2 + \dot{y}}$$

$$= \frac{P_x^2}{m} + \frac{P_y}{m}(P_y - eBx)$$

$$- \frac{P_x^2}{2m} - \frac{1}{2m}(P_y - eBx)^2$$

$$- \frac{eBx}{m}(P_y - eBx)$$

$$\boxed{H_2 = \frac{P_x^2}{2m} + \frac{(P_y - eBx)^2}{2m}} \qquad (17)$$

(g) Equations of motion from Hamiltonian H_1

$$\dot{x} = \frac{\partial H_1}{\partial P_x} \Rightarrow \dot{x} = \frac{1}{m}\left(P_x + \frac{eB}{2} y\right)$$

$$\dot{y} = \frac{\partial H_1}{\partial P_y} \Rightarrow \dot{y} = \frac{1}{m}\left(P_x - \frac{eB}{2} x\right) \qquad (18)$$

which are same as equ (9)

$$\dot{P}_x = -\frac{\partial H_1}{\partial x} \Rightarrow \frac{dP_x}{dt} = \frac{eB}{2m}\left(P_x - \frac{eB}{2} x\right)$$

which is same as (12), after using (9) or (18)

Similarly for $\frac{dP_y}{dt}$

(b) Note

$$L_1 - L_2 = -\frac{eB}{2}(x\dot{y} + y\dot{x})$$

$$= \frac{d}{dt}\left(-\frac{eB}{2}xy\right) \qquad (19)$$

which is a total derivative –

This explains why we get the same dynamics

This is related to the fact that the Lagrangian involves the vector potential rather than the field strength. The term is

$$e\,\vec{V}\cdot\vec{A} \qquad (20)$$

where $\vec{B} = \vec{\nabla}\times\vec{A}$

In our case

$$B = \partial_x A_y - \partial_y A_x \qquad (21)$$

~~In ccmos~~ If we use a different vector potential $\vec{A}'$

$$\vec{A}' = \vec{A} + \vec{\nabla}\alpha \qquad (22)$$

α being some function, we get the same $\vec{B}$

This corresponds to a gauge transformation.

L_1 ~~[illegible]~~ is the Lagrangian in the gauge

$$A_x^{(1)} = -\frac{eB}{2} y \qquad (23)$$
$$A_y^{(1)} = \frac{eB}{2} x$$

where L_2 is the Lagrangian in the gauge

$$A_y^{(2)} = eBx \qquad (24)$$
$$A_x^{(2)} = 0$$

They are related by a gauge transformation

$$A_i^{(2)} = A_i^{(1)} + \partial_i \left(\frac{eBxy}{2} \right) \qquad (25)$$

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Problem 2 (2.7.1)

(i) $$\{\omega,\lambda\} = \frac{\partial\omega}{\partial q_i}\frac{\partial\lambda}{\partial p_i} - \frac{\partial\lambda}{\partial q_i}\frac{\partial\omega}{\partial p_i}$$
$$= -\left(\frac{\partial\lambda}{\partial q_i}\frac{\partial\omega}{\partial p_i} - \frac{\partial\omega}{\partial q_i}\frac{\partial\lambda}{\partial p_i}\right)$$
$$= -\{\lambda,\omega\}$$

(ii) $$\{\omega,\lambda+\sigma\} = \frac{\partial\omega}{\partial q_i}\frac{\partial}{\partial p_i}(\lambda+\sigma) - \frac{\partial\omega}{\partial p_i}\frac{\partial}{\partial q_i}(\lambda+\sigma)$$
$$= \{\omega,\lambda\} + \{\omega,\sigma\}$$

(iii) $$\{\omega,\lambda\sigma\} = \frac{\partial\omega}{\partial q_i}\left(\lambda\frac{\partial\sigma}{\partial p_i} + \frac{\partial\lambda}{\partial p_i}\sigma\right)$$
$$-\frac{\partial\omega}{\partial p_i}\left(\lambda\frac{\partial\sigma}{\partial q_i} + \frac{\partial\lambda}{\partial q_i}\sigma\right)$$
$$= \lambda\left(\frac{\partial\omega}{\partial q_i}\frac{\partial\sigma}{\partial p_i} - \frac{\partial\omega}{\partial p_i}\frac{\partial\sigma}{\partial q_i}\right)$$
$$+\left(\frac{\partial\omega}{\partial q_i}\frac{\partial\lambda}{\partial p_i} - \frac{\partial\omega}{\partial p_i}\frac{\partial\lambda}{\partial q_i}\right)\sigma$$
$$= \lambda\{\omega,\sigma\} + \{\omega,\lambda\}\sigma$$

Problem 3 : (2.7.4)

$$\{\bar{x}, \bar{y}\} = \{x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta\}$$

$$= \sin\theta\cos\theta\,\{x,x\} - \sin^2\theta\,\{y,x\}$$
$$+ \cos^2\theta\,\{x,y\} - \sin\theta\,\{y,y\}$$

$$= (\cos^2\theta + \sin^2\theta)\,\{x,y\}$$
$$= 0$$

Since $\{x,y\} = 0$

$$\{\bar{x}, \bar{P}_x\} = \{x\cos\theta - y\sin\theta,\ P_x\cos\theta - P_y\sin\theta\}$$

$$= \cos^2\theta\,\{x, P_x\} - \sin\theta\cos\theta\,\{y, P_x\}$$
$$- \sin\theta\cos\theta\,\{x, P_y\} + \sin^2\theta\,\{y, P_y\}$$

$$= \cos^2\theta + \sin^2\theta = 1$$

Similarly $\{\bar{x}, \bar{P}_y\} = 0$

$$\{\bar{y}, \bar{P}_y\} = 1$$

$$\{\bar{y}, \bar{P}_x\} = 0$$

This proves that the transformation is canonical

Problem ~~4~~ (2.7.8)

(1) Since $q_i = q_i(\overline{q}_j)$

$$\dot{q}_i = \frac{dq_i}{dt} = \frac{\partial q_i}{\partial \overline{q}_j}\frac{d\overline{q}_j}{dt}$$

(2) $\Rightarrow \dfrac{\partial \dot{q}_i}{\partial \dot{\overline{q}}_j} = \dfrac{\partial q_i}{\partial \overline{q}_j}$

(3) By definition

$$\overline{p}_i = \left[\frac{\partial \mathcal{L}(\overline{q}, \dot{\overline{q}})}{\partial \dot{\overline{q}}_i}\right]_{\overline{q}_i}$$

However $\mathcal{L}$ is by definition

$$\mathcal{L}(\overline{q}, \dot{\overline{q}}) = \mathcal{L}(q, \dot{q})$$

Thus

$$\overline{p}_i = \left[\frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{\overline{q}}_i}\right]_{\overline{q}_i}$$

$$= \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\frac{\partial \dot{q}_j}{\partial \dot{\overline{q}}_i}\right]_{\overline{q}_i} = \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\right)_{\overline{q}}\left(\frac{\partial q_j}{\partial \overline{q}_i}\right)$$

However since $\overline{q} = \overline{q}(q)$, holding $\overline{q}$ constant is same as holding q constant. Thus

$$\overline{p}_i = \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\right)_q \frac{\partial q_j}{\partial \overline{q}_i} = \frac{\partial q_j}{\partial \overline{q}_i} p_j$$

Problem 5: (2.8.2)

If the generator is $g(P_i, q_i)$

$$\delta P_i = \varepsilon \{P_i, g\} = - \varepsilon \frac{\partial g}{\partial q_i} \quad (26)$$

$$\delta q_i = \varepsilon \{q_i, g\} = \varepsilon \frac{\partial g}{\partial q_i}$$

Then

$$\{q_i', P_j'\}$$
$$= \{q_i + \delta q_i, P_j + \delta P_j\}$$
$$= \{q_i, P_j\} + \varepsilon \{\{q_i, g\}, P_j\} \quad (27)$$
$$+ \varepsilon \{q_i, \{P_j, g\}\}$$

Now use the Jacobi identity

$$0 = \{q_i, \{P_j, g\}\} + \{P_j, \{g, q_i\}\} + \{g, \{q_i, P_j\}\} \quad (28)$$

Then use antisymmetry of Poisson bracket and $\{q_i, P_j\} = \delta_{ij}$ to get

$$0 = \{q_i, \{p_j, g\}\} + \{\{q_i, g\}, p_j\} + \{g, \delta_{ij}\}.$$

However $\{g, \delta_{ij}\} = 0$

Thus

$$0 = \{q_i, \{p_j, g\}\} + \{\{q_i, g\}, p_j\} \quad (29)$$

Use this in eqn (27) to get

$$\{q_i', p_j'\} = \{q_i, p_j\} = \delta_{ij}$$

~~Thus the~~ One may similarly verify that

$$\{q_i', q_j'\} = \{p_i', p_j'\} = 0$$

Thus transformation is canonical

Problem 6 (2.83)

The infinitesimal transformations for such a transformation are

$$\delta x = -\varepsilon y$$
$$\delta p_x = 0$$
$$\delta y = \varepsilon x$$
$$\delta p_y = 0$$

~~Suppose there is a g such that~~

$$\cancel{\delta H = \varepsilon \{H, g\}}$$

Then

$$\{x + \delta x, \; p_x + \delta p_x\}$$

$$= \{x - \varepsilon y, \; p_x\} = 1$$

$$\{x + \delta x, \; p_y + \delta p_y\}$$
$$= \{x - \varepsilon y, \; p_y\} = -\varepsilon$$

Thus $\{x', p_y'\} \neq 0$. Hence the transformation is not canonical

Suppose there is a $g(x, y; p_x, p_y)$

such that

$$\delta H = \varepsilon \{H, g\}$$

$$\{H, g\} = \frac{\partial H}{\partial x}\frac{\partial g}{\partial p_x} + \frac{\partial H}{\partial y}\frac{\partial g}{\partial p_y} - \frac{\partial H}{\partial p_x}\frac{\partial g}{\partial x} - \frac{\partial H}{\partial p_y}\frac{\partial g}{\partial y}$$

$$= m\omega^2\left(x\frac{\partial g}{\partial p_x} + y\frac{\partial g}{\partial p_y}\right) - \frac{1}{m}\left(p_x\frac{\partial g}{\partial x} + p_y\frac{\partial g}{\partial y}\right)$$

The first set of terms which multiply $m\omega^2$ comes from variation of H due to variation of x, the second set from variation of p.

Since $\delta p_x = \delta p_y = 0$ we must have $g(p_x, p_y)$ only. Since $\delta H = 0$ we must have

$$x\frac{\partial g}{\partial p_x} + y\frac{\partial g}{\partial p_y} = 0$$

The only solution of this equation is

$$g = \text{constant}.$$

Problem 7 (2.8.4)

For infinitesimal rotations in (x, p) plane

$$\delta x = \varepsilon p$$
$$\delta p = -\varepsilon x$$

This is clearly a canonical transformation

$$\{x', p'\} = \{x, p\} + \varepsilon\{\delta x, p\} + \varepsilon\{x, \delta p\} + O(\varepsilon^2)$$
$$= \{x, p\} = 1$$

For finite transformations

$$x' = \cos\theta \cdot x + \sin\theta \, p$$
$$p' = -\sin\theta \, x + \cos\theta \, p$$

$$\{x', p'\} = \cos^2\theta \{x, p\} - \sin^2\theta \{p, x\}$$
$$= \cos^2\theta + \sin^2\theta$$
$$= 1$$

so that this is canonical

If $g(x, p)$ is the generator

$$\delta x = \epsilon\{x, g\} = \varepsilon \frac{\partial g}{\partial p} = \varepsilon p \Rightarrow \frac{\partial g}{\partial p} = p$$
$$\delta p = \varepsilon\{p, g\} = -\varepsilon \frac{\partial g}{\partial x} = -\varepsilon x \Rightarrow \frac{\partial g}{\partial x} = x$$

The soln of these 2 equations is

$$g(x, p) = \frac{1}{2}(p^2 + x^2)$$

ie the Hamiltonian itself.

In fact the canonical transformation is just time evolution with $\theta = t$

Problem 8: (2.8.7)

The total energy can be easily calculated to be

$$E = \frac{1}{2} m\omega^2 (A^2 + B^2)$$

Imposing the initial and final conditions

$$x(0) = x_1 = A$$
$$x(T) = x_2 = A \cos\omega T + B \sin\omega T$$

Thus
$$A = x_1$$
$$B = \frac{x_2 - x_1 \cos\omega T}{\sin\omega T}$$

~~So that~~ The action is given by

$$S = \int_0^T dt \left[\frac{1}{2} m \dot{x}^2 - \frac{1}{2} m\omega^2 x^2 \right]$$

Substituting for A, B and evaluating the integral ~~in~~ we get the answer easily